

# An Asplund Ostrovskii quojection without Banach subspaces

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## Source problem

The source paper is J. Kakol and A. Leiderman, *Asplund locally convex spaces and the finest locally convex topology*, arXiv:2412.11510. In Section 4, after proving their quojection criterion for the Asplund property, they ask:

Does there exist a quojection Frechet lcs  $E$  which is Asplund not being a countable product of Banach spaces and yet  $E$  contains no infinite-dimensional Banach subspaces?

We prove that the answer is affirmative. In fact, Ostrovskii's original quojection without infinite-dimensional Banach subspaces already has an Asplund defining spectrum.

## Background

We use three standard facts about Banach-space Asplundness.

1. A separable Banach space is Asplund if and only if its dual is separable.
2. The class of Asplund Banach spaces is a three-space class: if  $Y$  and  $X/Y$  are Asplund, then  $X$  is Asplund.
3. Separability is also a three-space property for Banach spaces.

We also use the following theorem from the source paper.

**Theorem 1** (Kakol–Leiderman). *Let  $E = s\text{-proj}(E_n, P_n)$  be a quojection, i.e. a strict projective limit of Banach spaces. Then  $E$  is Asplund if and only if every Banach space  $E_n$  is Asplund.*

Ostrovskii's 1994 construction uses the following Bellenot–James space. Let  $1 < q < \infty$ , let  $Z$  be a separable Banach space, and let  $(Z_i)_{i=0}^\infty$  be an increasing sequence of finite-dimensional subspaces with  $Z_0 = \{0\}$  and dense union in  $Z$ . For finitely supported sequences  $z = (z_i)$ ,  $z_i \in Z_i$ , define

$$2\|z\|_{J,q}^q = \sup \left\{ \sum_{i=1}^{k-1} \|z_{\eta(i)} - z_{\eta(i+1)}\|^q + \|z_{\eta(k)}\|^q \right\},$$

where the supremum is taken over all finite increasing sequences  $0 \leq \eta(1) < \dots < \eta(k)$ . Let  $J(q, Z)$  denote any completion obtained in this way.

Ostrovskii records the key identity

$$J(q, Z)^{**}/J(q, Z) \cong Z$$

isometrically.

## The Asplund check

**Lemma 1.** *Let  $Z$  be a separable Asplund Banach space and  $1 < q < \infty$ . For every Bellenot–James space  $X \in J(q, Z)$ , both  $X$  and  $X^{**}$  are Asplund.*

*Proof.* The space  $X$  is separable, because it is the completion of the finitely supported sequences over a countable sequence of finite-dimensional spaces. By Ostrovskii’s quoted form of Bellenot’s theorem,

$$X^{**}/X \cong Z.$$

Since both  $X$  and  $Z$  are separable, the middle space  $X^{**}$  is separable by the three-space property of separability.

Now  $X^{**}$  is the dual of  $X^*$ . A Banach space whose dual is separable is separable, hence  $X^*$  is separable. Since  $X$  is separable with separable dual,  $X$  is Asplund.

Finally,  $X$  is an Asplund subspace of  $X^{**}$  and  $X^{**}/X \cong Z$  is Asplund. By the three-space property for Asplund Banach spaces,  $X^{**}$  is Asplund.  $\square$

## The construction

**Theorem 2.** *There exists a quojection Frechet locally convex space  $E$  which is Asplund, is not isomorphic to a countable product of Banach spaces, and contains no infinite-dimensional Banach subspace.*

*Proof.* Choose a strictly decreasing sequence  $q(1) > q(2) > \dots$  in  $(1, \infty)$ . Ostrovskii defines Banach spaces  $F_n$  recursively as follows. Put

$$F_1 = \ell_{q(1)}.$$

For  $n \geq 2$ , choose  $X \in J(q(n), F_{n-1})$  and set

$$F_n = X^{**}.$$

The quotient identity  $X^{**}/X \cong F_{n-1}$  gives a quotient map

$$R_{n-1} : F_n \longrightarrow F_{n-1}.$$

Thus  $(F_n, R_n)$  is a projective sequence with surjective bonding maps.

Ostrovskii proves that these  $F_n$  satisfy the Metafuno–Moscatelli criterion:

$$j \neq k \implies F_j \text{ and } F_k \text{ have no isomorphic infinite-dimensional subspaces.}$$

Therefore the projective limit

$$E = \text{proj}(F_n, R_n)$$

is a twisted quojection, i.e. it is not isomorphic to a countable product of Banach spaces, and it contains no infinite-dimensional Banach subspaces.

It remains only to check the Asplund property. The space  $F_1 = \ell_{q(1)}$  is reflexive, hence separable Asplund. If  $F_{n-1}$  is separable Asplund, then Lemma 1 applied to  $Z = F_{n-1}$  gives that  $F_n = X^{**}$  is Asplund; it is also separable by the proof of the lemma. Induction shows that every  $F_n$  is Asplund.

By Theorem 1, the quojection  $E$  is Asplund. This  $E$  has all three required properties.  $\square$

## Relation to the source paper

The source paper already points out that Ostrovskii’s mechanism would solve the problem if the Banach spaces in the defining sequence could be chosen Asplund. The observation above is that no new construction is needed: Ostrovskii’s recursively defined spaces are Asplund by a short induction using  $X^{**}/X \cong Z$ , separability, and the three-space property for Asplund Banach spaces.

## Novelty and literature check

Local run indexes were searched for arXiv:2412.11510, the source title, Asplund spaces, finest locally convex topology, quojection, Ostrovskii, and the exact final problem phrase. No existing packet or attempt for this target was found.

Web searches on 2026-06-19 for the source title, the final quojection problem, and Ostrovskii’s *Quojections without Banach subspaces* found the two papers used here, but no explicit published answer to the Asplund-strengthened question.

## References

- J. Kakol and A. Leiderman, *Asplund locally convex spaces and the finest locally convex topology*, arXiv:2412.11510.
- M. I. Ostrovskii, *Quojections without Banach subspaces*, arXiv:math/9403209; Bull. Polish Acad. Sci. **44** (1996), 143–146.
- R. Eyland and B. Sharp, *A factor theorem for locally convex differentiability spaces*, Bull. Austral. Math. Soc. **43** (1991), 101–113.
- J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I: Sequence Spaces*, Springer, 1977.

## Human review recommendation

The main point for review is Lemma 1. It uses only standard permanence facts, but it is the step not spelled out in either source paper. The rest of the argument is exactly Ostrovskii’s projective-spectrum criterion combined with Kakol–Leiderman’s quojection-Asplund theorem.